

REAL ANALYSIS GENERAL EXAM FALL 2022

Solve as many problems as you can. Full solutions on a smaller number of problems will be worth more than partial solutions on several problems.

Problem 1.

Compute

$$\lim_{n \rightarrow \infty} \int_0^\infty \frac{n \sin(x/n)}{x(1+x^2)} dx$$

Problem 2.

Fix $a < b$ in $\mathbb{R}$. Recall that $h : [a, b] \rightarrow \mathbb{C}$ is absolutely continuous if for every $\varepsilon > 0$ there is a $\delta > 0$ so that if $((a_j, b_j))_{j=1}^k$ are disjoint intervals in $[a, b]$ with $\sum_{j=1}^k (b_j - a_j) < \delta$, then $\sum_{j=1}^k (f(b_j) - f(a_j)) < \varepsilon$. For a Lipschitz function $g : [a, b] \rightarrow \mathbb{C}$ we set

$$\|g\|_{Lip} = \sup_{x \neq y, x, y \in [a, b]} \frac{|g(x) - g(y)|}{|x - y|}$$

(a) Show that $f : [a, b] \rightarrow \mathbb{C}$ is Lipschitz if and only if f is absolutely continuous and $f' \in L^\infty([a, b])$. (b) If $f : [a, b] \rightarrow \mathbb{C}$ is Lipschitz, show that $\|f\|_{Lip} = \|f'\|_\infty$.

Problem 3.

Let (X, μ) be a σ -finite measure space. Show that if $f, g \in L^1(X, \mu)$ with $0 \leq f, g$ a.e., then

$$\|f - g\|_1 = \int_0^\infty \mu(\{x : f(x) > t\} \Delta \{x : g(x) > t\}) dt$$

Here $E \Delta F = E \setminus F \cup F \setminus E$ for sets $E, F \subseteq X$. Suggestion: it might be helpful to first show that for $a, b \in [0, \infty)$ we have

$$|a - b| = \int_0^\infty |1_{(t, \infty)}(a) - 1_{(t, \infty)}(b)| dt$$

Note: for this problem you may take for granted that the function $X \times (0, \infty) \rightarrow \{0, 1\}$ given by $(y, t) \mapsto 1_{\{x : f(x) > t\}}(y)$ and that the function $t \mapsto \mu(\{x : f(x) > t\} \Delta \{x : g(x) > t\})$ are measurable functions.

Problem 4.

Let (X, Σ) be a measurable space. Recall that if η is a signed measure on Σ , then $|\eta| = \eta_1 + \eta_2$ where η_1, η_2 are the unique nonnegative measures with $\eta = \eta_1 - \eta_2$ and $\eta_1 \perp \eta_2$. Further, $\|\eta\|_{TV} = |\eta|(X)$. Suppose that μ, ν are signed measures on Σ , that $\|\mu\|_{TV}, \|\nu\|_{TV} < +\infty$ and that $|\mu|, |\nu|$ are mutually singular. (a) If $\mu = \mu_1 - \mu_2, \nu = \nu_1 - \nu_2$ with μ_i, ν_j nonnegative measures and $\mu_1 \perp \mu_2, \nu_1 \perp \nu_2$, show that $\mu_i \perp \nu_j$ for all $i, j \in \{1, 2\}$. (b) Show that

$$\|\mu + \nu\|_{TV} = \|\mu\|_{TV} + \|\nu\|_{TV}$$

Problem 5.

(a) For $f \in L^1([0, 1])$, set L_f be the set of $x \in [0, 1]$ so that

$$\lim_{r \rightarrow 0} \frac{1}{2r} \int_{(x-r, x+r)} |f(y) - f(x)| dy = 0$$

State the conclusion of the Lebesgue's differentiation theorem for L_f . (b) For $n \in \mathbb{N}$, and $0 \leq j \leq 2^n - 1$, set $I_{n,j} = [j2^{-n}, (j+1)2^{-n})$. For $f \in L^1([0, 1])$, define

$$E_n f = \sum_{j=0}^{2^n-1} \left(\frac{1}{m(I_{n,j})} \int_{I_{n,j}} f(t) dt \right) 1_{I_{n,j}}$$

Show that

$$\lim_{n \rightarrow \infty} (E_n f)(x) = f(x) \text{ for almost every } x \in [0, 1]$$